



DATA ANALYTICS ROADMAP



— Step by Step to Become a Data Analyst —

★ Learn. Practice. Analyze. Inspire with Data.

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1. Foundations

- What is Data Analytics?
- Types of Data
- Data Lifecycle
- Basic Statistics (Mean, Median, Mode, Std Dev)

🕒 1-2 Weeks

YOU WILL BE ABLE TO:
Understand key concepts of data and analytics.



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2. Excel Essentials

- Formulas & Functions (SUM, IF, VLOOKUP)
- Data Cleaning in Excel
- Pivot Tables & Charts
- Data Visualization Basics

🕒 1-2 Weeks

YOU WILL BE ABLE TO:
Work with data and create insightful reports in Excel.



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3. SQL for Data Analysis

- SELECT, WHERE, ORDER BY
- Aggregations (SUM, AVG, COUNT)
- GROUP BY, HAVING
- JOINS (INNER, LEFT, RIGHT)
- Subqueries

🕒 2-3 Weeks

YOU WILL BE ABLE TO:
Extract, manipulate and analyze data using SQL.



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4. Data Cleaning & Preparation

- Handling Missing Values
- Removing Duplicates
- Data Transformation
- Outlier Detection

🕒 1-2 Weeks

YOU WILL BE ABLE TO:
Clean and prepare data for accurate analysis.



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5. Python for Data Analysis

- Python Basics (Variables, Loops, Functions)
- Libraries: NumPy, Pandas
- Data Manipulation with Pandas
- Descriptive Statistics

🕒 2-3 Weeks

YOU WILL BE ABLE TO:
Analyze and manipulate data using Python.



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6. Data Visualization

- Data Visualization Principles
- Matplotlib & Seaborn (Python)
- Charts: Line, Bar, Pie, Histogram, Boxplot
- Dashboards with Plotly / Power BI Basics

🕒 2-3 Weeks

YOU WILL BE ABLE TO:
Create clear and effective visualizations.



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7. Exploratory Data Analysis (EDA)

- Univariate, Bivariate, Multivariate Analysis
- Correlation Analysis
- Identifying Patterns & Trends

🕒 1-2 Weeks

YOU WILL BE ABLE TO:
Explore data and find meaningful insights.



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8. Data Analysis & Reporting

- KPIs and Metrics
- Business Analysis Techniques
- Storytelling with Data
- Reports & Dashboards (Excel / Power BI)

🕒 2-3 Weeks

YOU WILL BE ABLE TO:
Create reports and dashboards that drive decisions.



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9. Advanced Topics (Optional)

- Advanced SQL
- Advanced Excel (Power Query, Power Pivot)
- Tableau / Power BI Advanced
- Basics of Machine Learning for Analysts

🕒 2-4 Weeks

YOU WILL BE ABLE TO:
Handle complex data and advanced analysis.



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10. Projects & Portfolio

- Work on Real-world Projects
- Case Studies
- Build a Portfolio and Showcase
- Share on GitHub / LinkedIn

🕒 Ongoing

YOU WILL BE ABLE TO:
Apply your skills to real problems and get noticed.



TOOLS YOU SHOULD KNOW



Excel



SQL



Python



Power BI /
Tableau



Pandas

PROJECT IDEAS

- ✓ Sales Analysis Dashboard
- ✓ Customer Segmentation
- ✓ E-commerce Analysis
- ✓ Marketing Campaign Analysis
- ✓ HR Analytics Dashboard

TIPS FOR SUCCESS

- ✓ Practice Every Day
- ✓ Work on Real Data
- ✓ Build Projects
- ✓ Ask Questions & Keep Learning
- ✓ Be Curious & Stay Consistent

★ Consistency is the key! Keep learning, keep analyzing, keep growing. 📈